The Bus

The individual modules within an SoC need to communicate with each other. The system bus is what connects them. The bus is a set of wires and protocols. All modules attached to a bus should adhere to the wiring as well as the protocol. Without the bus, the modules cannot send or receive data. For example, AMBA protocol is a popular bus architecture designed by ARM systems.

Memory Controller

Let's say the processor needs to read "128 Bytes of data from address *0x00DEA0FE* from the RAM. While to us humans, that might look like a valid request, such a request is not enough to actually fetch the data. This is because the protocols to read ROMs are different than that needed to read from RAM which are again differ from protocols for reading from persistent storage units. To fulfil a read or write request, the low-level protocol for storing and retrieving data needs to be known. The Memory controller is the unit responsible for this work.

Interfaces

An SoC normally has multiple interface controllers depending on the use case. For example, a mobile phone SoC would contain interfaces for USB, Wi-Fi, LTE Radio, Bluetooth, Display (graphics), SPI (Serial Peripheral Interface) etc. These interfaces are connected to the processor via the system bus and are connected to their respective h/w units (e.g. a Bluetooth transceiver) as well.

Many of these units can be called SoCs in their own right. For example, a wireless connectivity system on a mobile can contain the transceiver hardware for Bluetooth, Wi-Fi, GSM/LTE as well as the DSP system for these technologies – all of this can be outside the main SoC.

Power Regulators

In an electronic or electrical system, everything depends on electricity and in a country like India, voltages are not guaranteed. While there are circuits outside of the SoC taking care of the power supply, an SoC needs to have a voltage and current regulator built right into it so that it can save the rest of the circuits from getting fried due to voltage spikes or causing an error when electrical energy is below the required ranges. They are responsible for maintaining voltages and current levels. These circuits can also be con-